MIL-L-40138D
31 December 1985
SUPERSEDING
MIL-L-40138C
30 September 1969

MILITARY SPECIFICATION

LOAD COUPLERS - 8 AND 12 SPOOL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and two sizes of load couplers for airdrop equipment.

1.2 Classification. Load couplers shall be two sizes, as specified (see 6.2):

Size 1 - 8 Spool
Size 2 - 12 Spool

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1670

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

DRAWING

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

11-1-66 - Load Coupler, 8 and 12 Spool; Assembly and Details

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

UNITED STATES OF AMERICA STANDARDS INSTITUTE

USAS B46.1-1965 - Surface Texture

(Application for copies should be addressed to United States of America Standards Institute, 10 East 40th Street, New York, NY 10016.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

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* 3.2 Materials, construction, and components. Material, construction, and components shall be as specified herein and on Drawing 11-1-66. Materials not definitely specified shall be of the quality normally used by the manufacturer, provided that the completed item complies with all provisions of this document. Recycled material may be used (see 6.4).

3.3 Identification marking. Marking for identification shall be as specified on Drawing 11-1-66.

3.4 Workmanship. The load couplers shall be clean and free from sharp edges, burrs, or slivers. Components shall not be cracked, fractured, or malformed. Surface roughness shall not exceed the limits specified on Drawing 11-1-66. Finish designations shall be interpreted in accordance with USAS B46.1-1965.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3 and 6.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), a completely fabricated load coupler shall be inspected for all end item inspection provisions in this document. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

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4.4.2 In-process inspection. Inspection shall be made of the load couplers during fabrication to determine conformance to the requirements of this document. The Government reserves the right to exclude from consideration for acceptance any materials for which in-process inspection has indicated nonconformance.

4.4.3 End item inspection. The load couplers shall be examined for visual characteristics and proper dimensions. The lot shall be expressed in units of load couplers of the same size. The sample unit shall be one load coupler.

4.4.3.1 Visual examination. The load couplers shall be examined for the defects listed in table I. The inspection level shall be II, and the acceptable quality level (AQL) shall be 1.0 for major defects and 4.0 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of visual defects

Examine	Defects	Classification	
		Major	Minor
Finish	Parts not finished as specified	101	
	Any area of rust or corrosion	102	
	Coating is flaking or otherwise not adherent	103	
	Any breaks, scratches, or tool marks through coating to metal		201
	Dirt, smudge, oil, grease, or foreign matter		202
	Color not as specified		203
Design	Any departure from specified requirements not otherwise classified herein	104	
Construction and workmanship	Any component cracked, fractured, or malformed	105	
	Any surface roughness exceeding specified limit		204
	Any sharp edges, burrs, or slivers	106	
Identification marking	Missing, incomplete, illegible, not permanently indented, or not as specified		205

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4.4.3.2 Dimensional inspection. Load couplers shall be inspected for compliance with dimensions and tolerances specified on the drawing. Any deviation from the specified dimensions or tolerances shall constitute a defect. The inspection level shall be S-2, and the AQL shall be 4.0, expressed in terms of defects per hundred units.

4.4.4 Packaging inspection. An examination shall be made to determine compliance with the applicable packing and marking requirements. The lot size shall be the number of shipping containers in the inspection lot. The sample unit shall be one shipping container fully packaged. Characteristics to be examined and the applicable defects are listed in table II. The inspection level shall be S-2, and the AQL for one class of load coupler shall be 6.5, expressed in terms of defects per hundred units.

TABLE II. Packaging inspection

Examine	Defect
Marking (container)	Omitted, incorrect, illegible, or of improper size, location, sequence, or method of application
Materials	Component missing, damaged, defective, or not as specified
Workmanship	Improperly positioned in container, improper closure of container, or loose strapping, bulging, or distortion of containers
Contents	Number more or less than specified

5. PACKAGING

5.1 Packing. Packing shall be level A.

5.1.1 Level A packing. Fifteen size 1 couplers (8 spool) or ten size 2 couplers (12 spool) completely assembled and with cotter pins spread open only enough to prevent their falling off the bolts during transit shall be packed in a snug-fitting shipping container conforming to overseas type, style A, type 3 load of PPP-B-601, or class 2, style 2 or 4, type 3 load of PPP-B-621. Size 1 couplers shall be positioned with bolt heads facing down, arranged five

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long, three wide, and one deep within each container. Size 2 couplers shall be positioned with the smaller plate dimensions facing up and nuts facing the side panels of the container, arranged five long, two wide, and one deep within each container. Closure and strapping shall be in accordance with the appendix of the applicable container document.

5.2 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The load coupler assemblies are intended for use as detachable load suspension links for connecting airdrop equipment to cargo parachutes.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Size required (see 1.2).
c. Whether first article inspection is required, and if required, number of samples (see 3.1 and 6.3).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

* 6.4 Recycled material. The contractor is encouraged to use recycled material when practical, provided that it meets the requirements of this document (see 3.2).

6.5 International standardization. The design provisions of this document and the applicable drawing are the subject of international standardization agreement ABC Air Standardization Agreement 44/20. When amendment, revision, or cancellation of this document is proposed that will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

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6.6 Changes from previous issue. The margins of this document are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-40138D	Load Couplers - 8 and 12 Spool

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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